# High-Performance Computing for Science in the Age of AI

Simulation, agents and machine learning on Argonne's supercomputers

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Argonne NATIONAL LABORATORY

U.S. DEPARTMENT of ENERGY

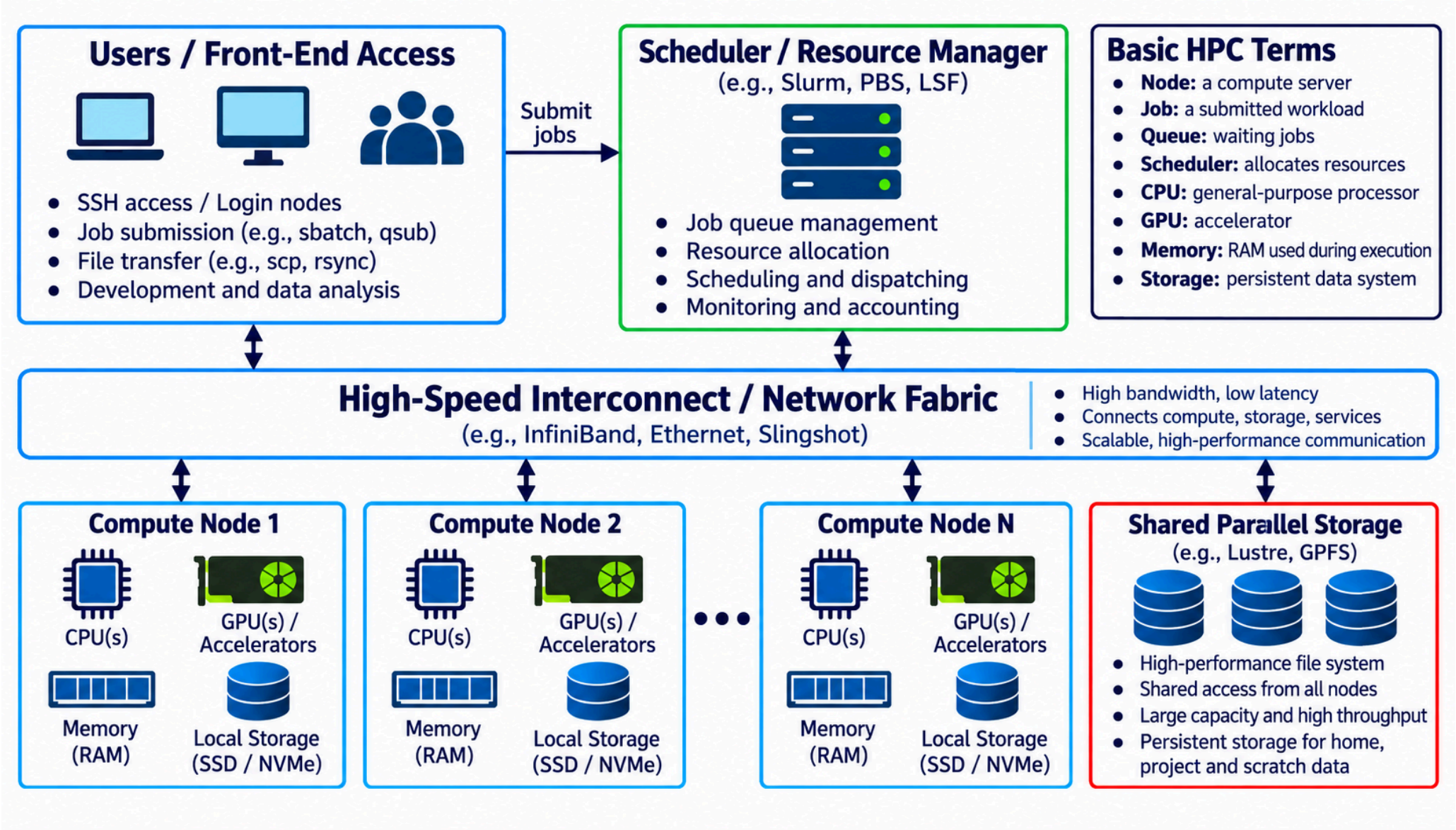
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# HPC

# What is HPC?

- High-Performance Computing (HPC) is the use of supercomputers to solve advanced computation problems
- Supercomputer is a tightly integrated system of many compute nodes connected by a high-speed network
- Co-design of hardware, system software, networking, and storage to achieve maximum performance
- From <1000 operations/second in the late 1940s to today's exascale computing (10^18 flops/s), supercomputer speed has steadily improved by about a factor of 200 times every decade.

# HPC Architecture



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# Argonne and HPC

# Argonne National Laboratory

**1946**
the first US national laboratory

**3,836**
employees, 769 grad/undergrad students

**4th**
Aurora, 4th world fastest supercomputer

**21**
research divisions, 6 national user facilities

**3**
Nobel laureates among its researchers

**250+**
open-source software tools and models

*anl.gov/our-history/timeline, anl.gov by the numbers FY2024, TOP500 June 2026*

# The Advanced Photon Source

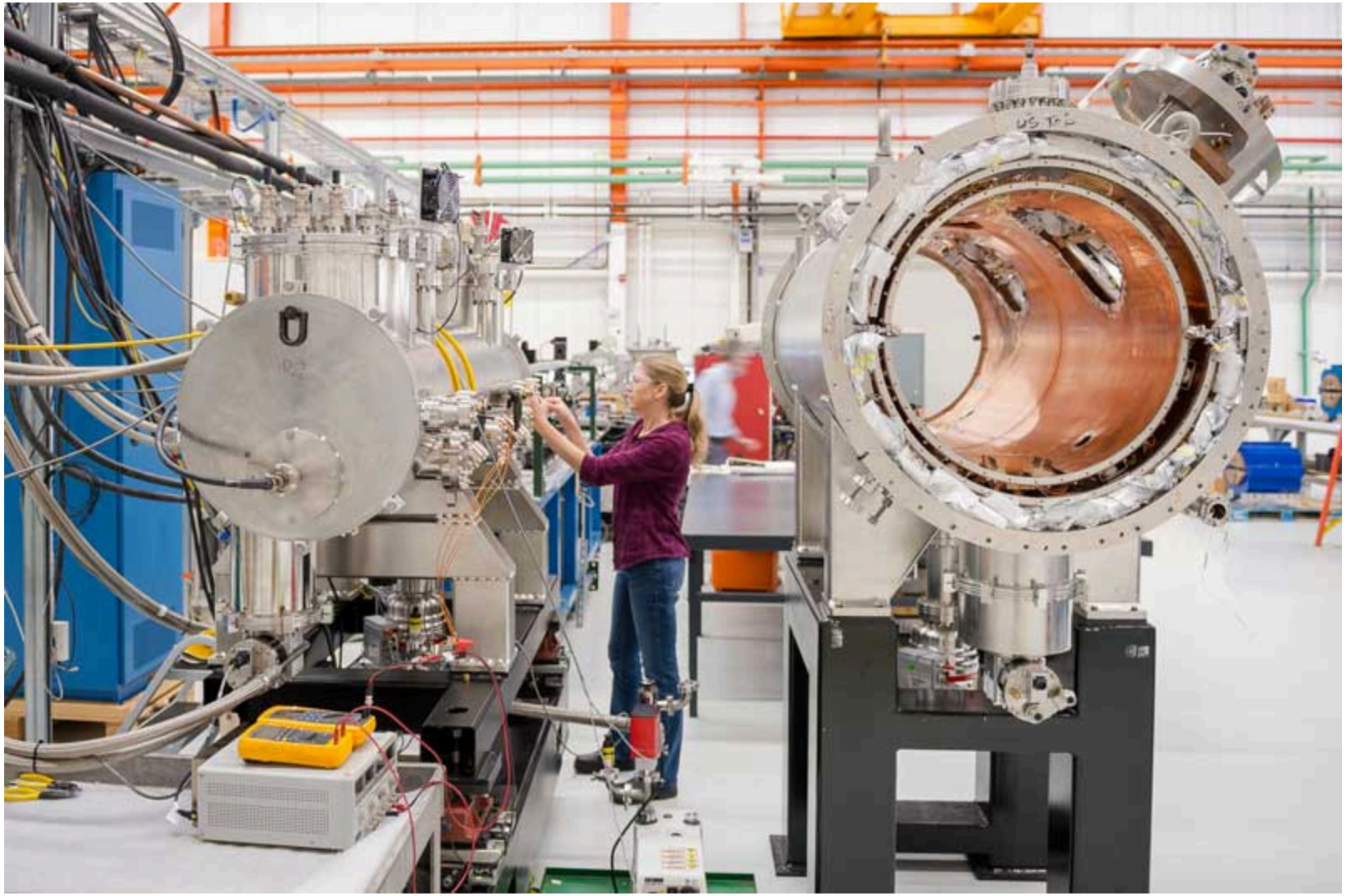


Brightest storage-ring X-ray source in the world

Ring is about 1.1 km around

*Advanced Photon Source, aps.anl.gov*

# 72 experiment stations around the ring



Imaging · spectroscopy · scattering · structural biology

*Advanced Photon Source · Image: Argonne National Laboratory*

# Computing, Environment and Life Sciences Directorate

**BIO**
Biosciences

**CPS**
Computational Science

**DSL**
Data Science and Learning

**EVS**
Environmental Science

**MCS**
Mathematics and Computer Science
MPICH, PETSc, runtimes, workflows

**ALCF**
Leadership Computing Facility
Aurora · Polaris · Sophia · AI Testbed

5 research divisions + 1 national user facility

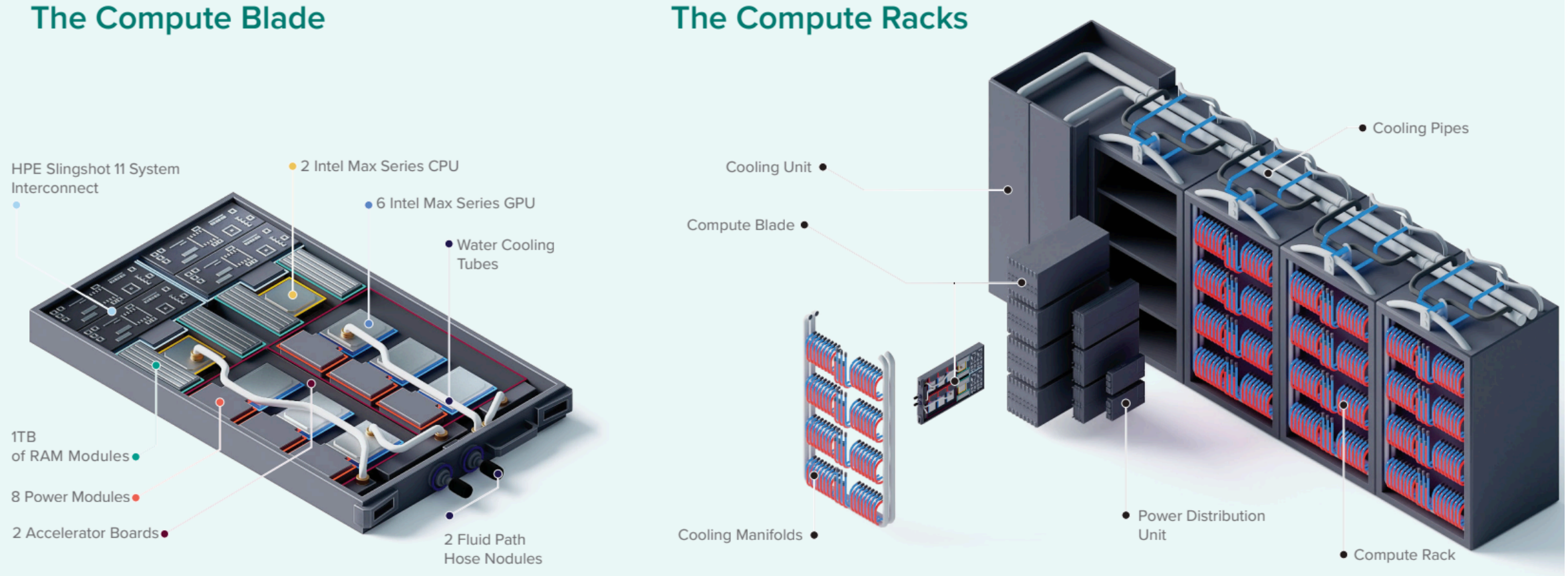
# Aurora




| 10,624 | 63,744 | 21,248 | 20.4 PB | exascale |
|---|---|---|---|---|
| COMPUTE NODES | INTEL GPUS | XEON CPUS | SYSTEM MEMORY | $10^{78}$ OPS / SECOND |

*ALCF Aurora · Image: Argonne National Laboratory*

# Aurora Architecture




>2 Exaflop Supercomputer (2 billion billion calculations/second)

*Image: ALCF Computing Resource (https://www.alcf.anl.gov/sites/default/files/2024-07/Aurora_FactSheet_2024.pdf)*

# Aurora Fun Facts



- 482 kilometers of networking cables
- 60 megewatts of power
- 166558 liters of water (closed loop)
- Weight 600 ton

*[Argonne lays the groundwork for its Aurora exascale supercomputer](https://www.alcf.anl.gov/news/argonne-lays-groundwork-its-aurora-exascale-supercomputer)*

# Polaris



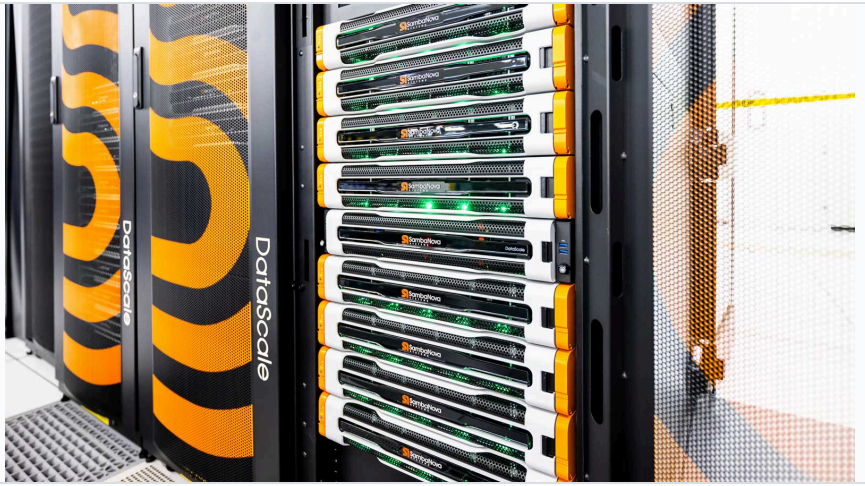

| 560 | 2,240 | 44 PF | HPE |
|---|---|---|---|
| COMPUTE NODES | NVIDIA A100 GPUS | TENSOR-CORE FP64 PEAK | APOLLO 6500 GEN10+ |

*ALCF Polaris · Image: Argonne National Laboratory*

# Advanced AI accelerators for scientific research




**Cerebras CS-3**, 4 nodes, 900,000 cores per accelerator, 125 PFLOPS FP16 each




**Metis (SambaNova SN40L)**, 32 accelerators; powers the public inference endpoints




**GroqRack**, 72 accelerators, >750 TFLOPS INT8 each, low-latency inference



**Graphcore Bow Pod64**, 64 IPUs, 1,472 processing units each

The **ALCF AI Testbed**, an infrastructure for the next-generation of AI-accelerator machines

# Upcoming next-generation AI supercomputing systems at Argonne




Solstice: 100,000 NVIDIA Blackwell GPUs, will be the largest AI supercomputer in the DOE's lab complex

Equinox: 10,000 NVIDIA Blackwell GPUs expected to be delivered in 2026

*DOE, Argonne, NVIDIA and Oracle partnership: https://www.anl.gov/article/argonne-expands-nations-ai-infrastructure-with-powerful-new-supercomputers*

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# Science with HPC

# How a scientist gets compute hour




**One node-hour = one whole compute node, held for one hour.** A job on 100 nodes for 3 hours spends 300 node-hours, whether or not every GPU was used. An Aurora node is 2 CPUs and 6 GPUs, so 1,000,000 node-hours is 6,000,000 GPU-hours.

# What a year of Aurora and Polaris is spent on

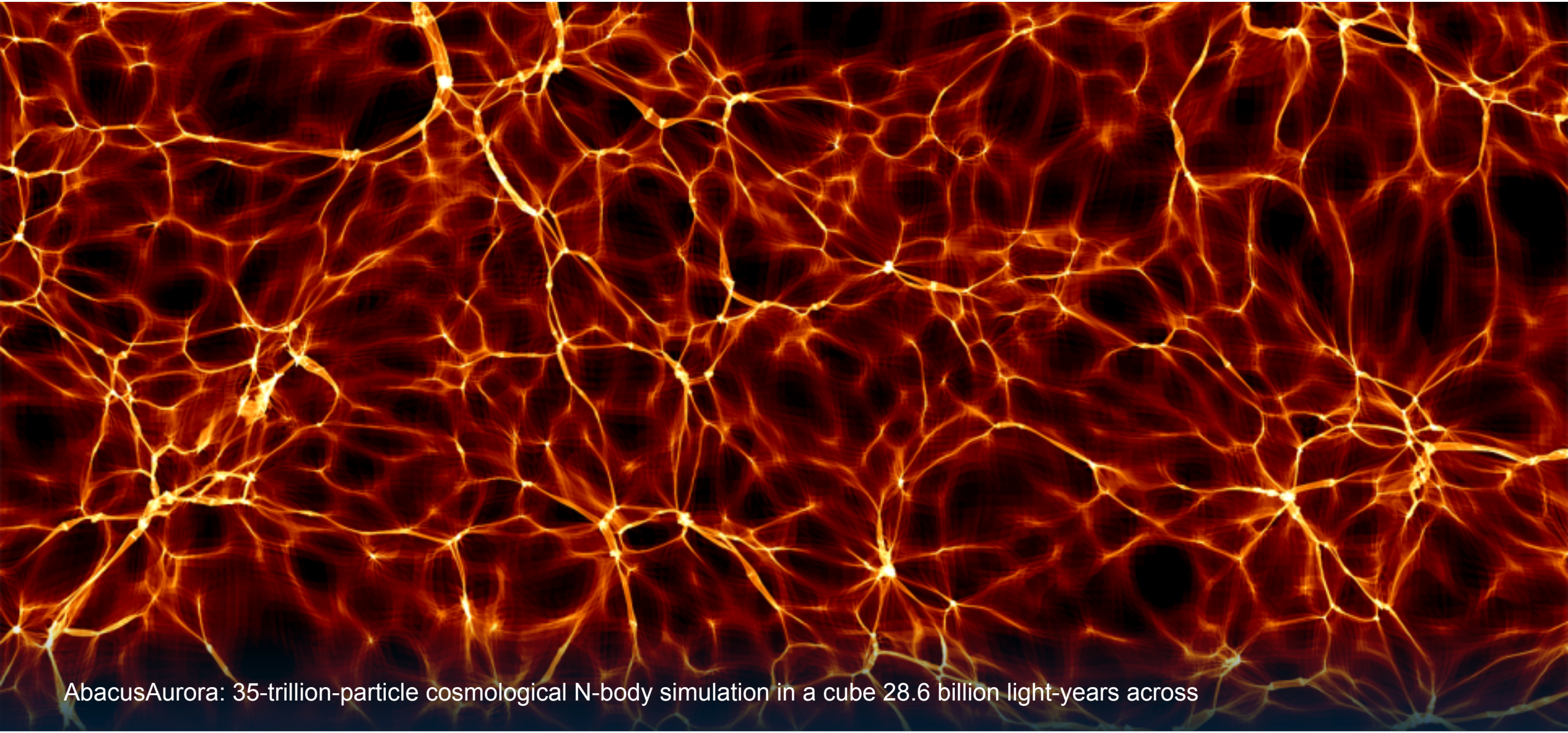
| DOMAIN | AWARD | PI · INSTITUTION | NODE-HOURS |
|---|---|---|---|
| **Earth Science** | E3SM, 200 m climate model | Aaron Donahue · Lawrence Livermore National Laboratory | **1,000,000 Aurora** |
| **Computer Science** | AuroraGPT foundation model | Rick Stevens · Argonne National Laboratory | **3,000,000 Aurora** |
| **Physics** | AbacusAurora, 35-trillion-particle cosmology | Daniel Eisenstein · Harvard University | **510,000 Aurora** |
| **Biological Sciences** | Protein design from language prompts | Rama Ranganathan · University of Chicago | **550,000 Aurora** |
| **Engineering** | Open fan / wing for sustainable aviation | Stephan Priebe · GE Aerospace Research | **400,000 Aurora** |
| **Materials Science** | Digital twins of catalysts and batteries | Boris Kozinsky · Harvard University | **430,000 Aurora** |
| **Energy Technologies** | AI-facilitated tokamak profile prediction | Robert Hager · Princeton Plasma Physics Laboratory | **1,000,000 Aurora** |

# The user community is international

| PROJECT | PI · INSTITUTION | NODE-HOURS |
|---|---|---|
| COMPBIO4, human digital twins for personalised medicine | Peter Coveney · University College London, UK | **500,000 Aurora** |
| Ab initio foundation models for RNA-targeted therapeutics | Jean-Philip Piquemal · Sorbonne University, France | **1,300,000 Aurora** |

**Open to any researcher or research organization in the world.** Proposals are judged on scientific merit and computational readiness by an international peer-review panel.

# 510,000 node-hours looks like this



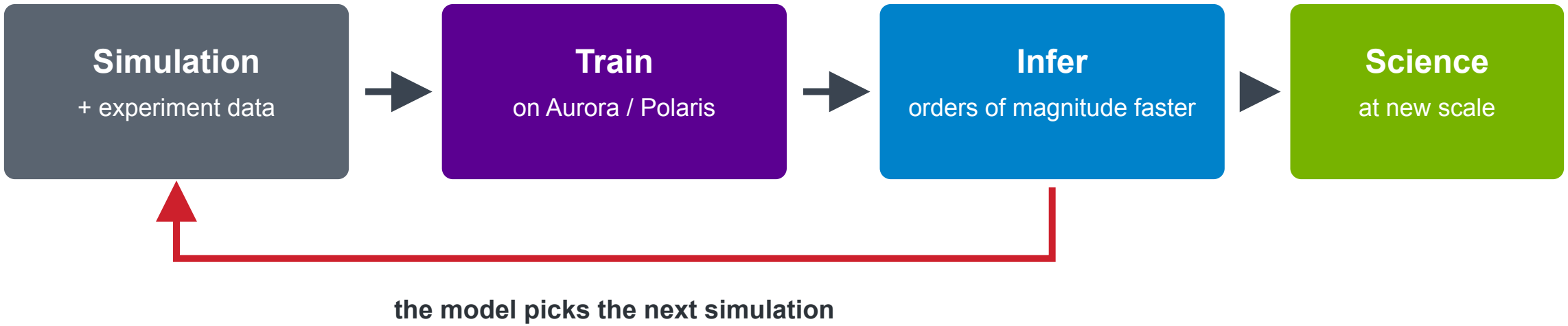
AbacusAurora: 35-trillion-particle cosmological N-body simulation in a cube 28.6 billion light-years across

*https://www.alcf.anl.gov/science/case-studies/simulating-universe-unprecedented-scale*

04

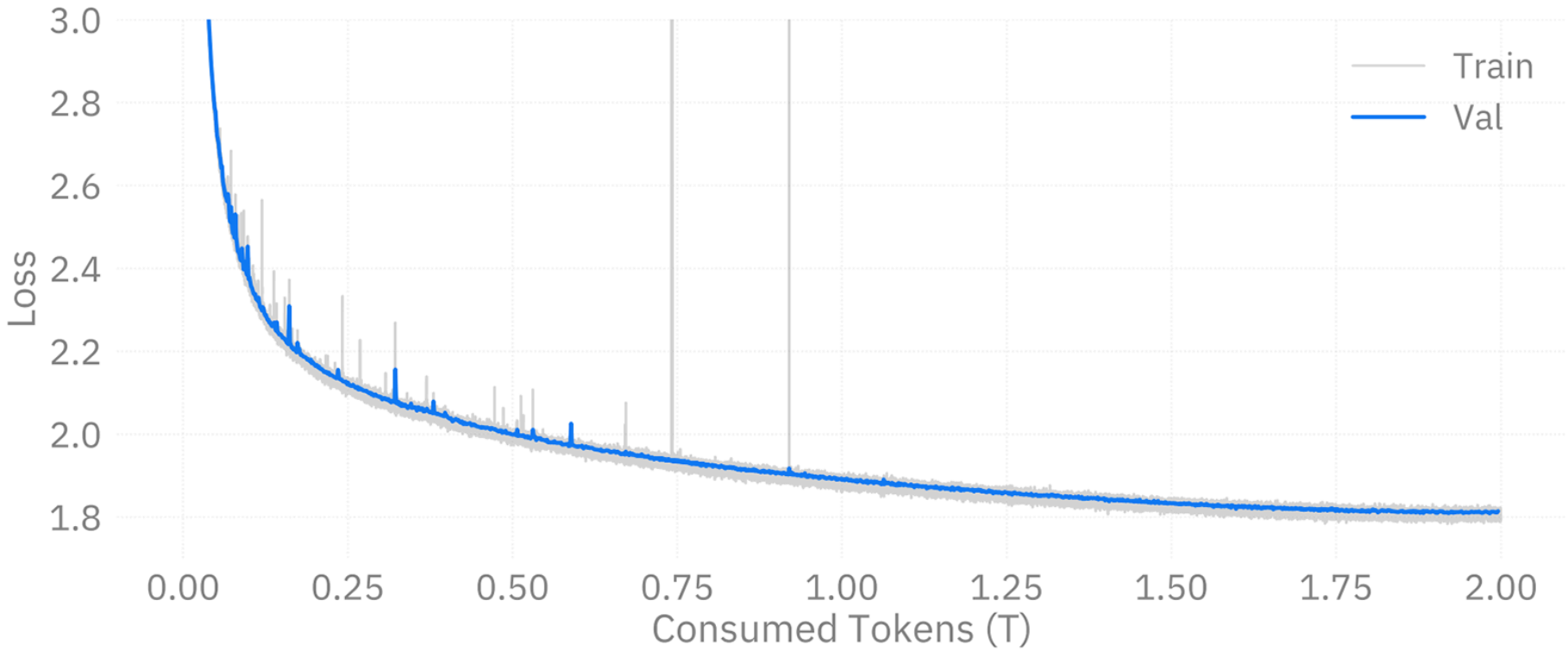
# AI for Science

# AI is now half of what these machines do




The machine is used twice: to make the training data, and then to run the model inside the workflow.

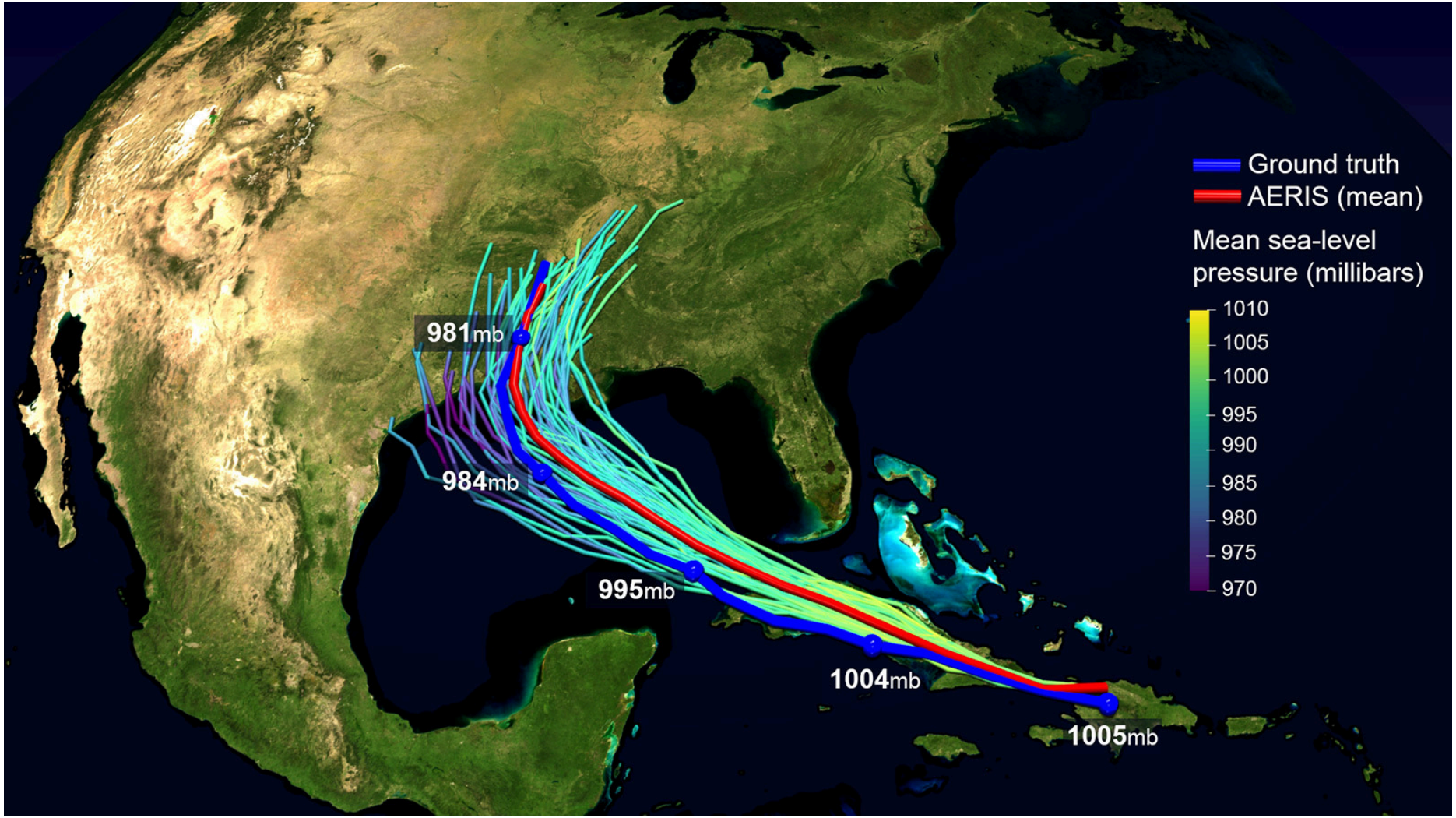
# AuroraGPT, a foundation model built for science




3,000,000 Aurora node-hours + 75,000 Polaris node-hours

Scientifically trained and aligned models, plus the tooling to put them in workflows

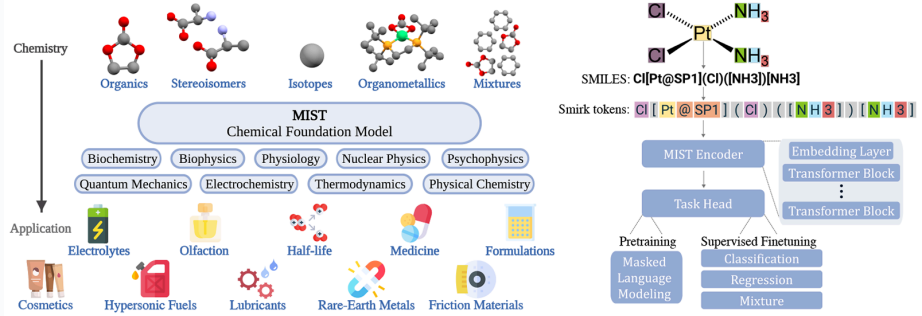
# AERIS, an AI Earth-system model



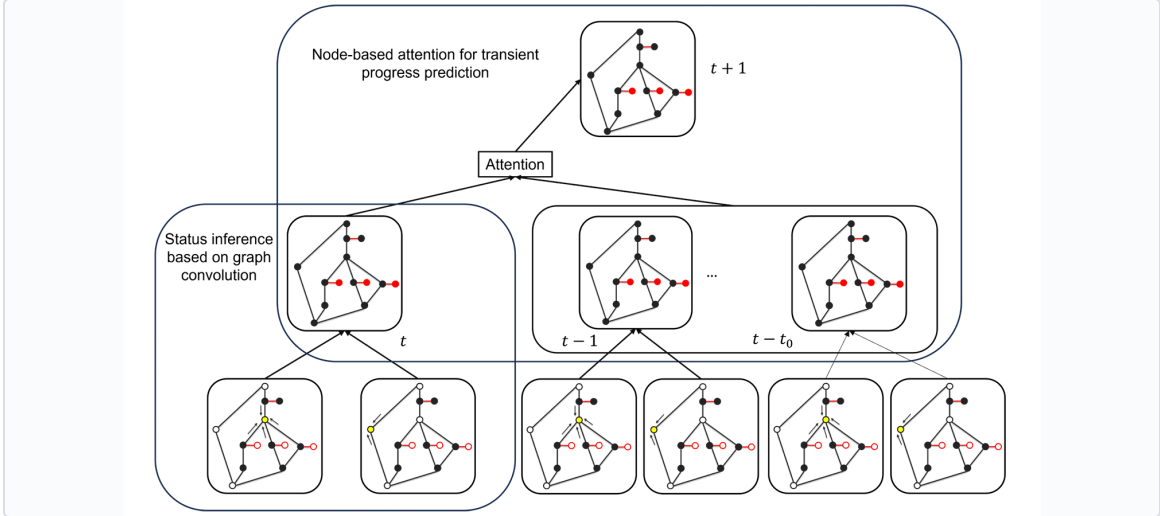

Hurricane Laura, forecast by a model trained on Aurora

Learned, not integrated: a forecast that took a supercomputer now takes a fraction of one

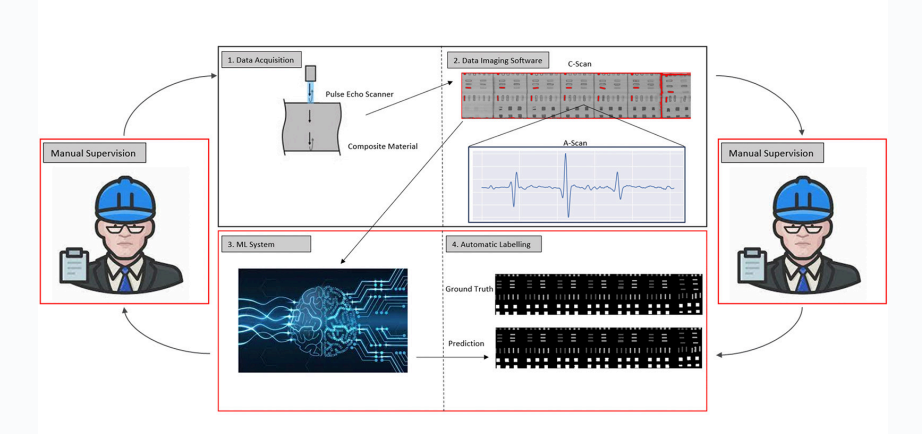
# More AI running on ALCF machines



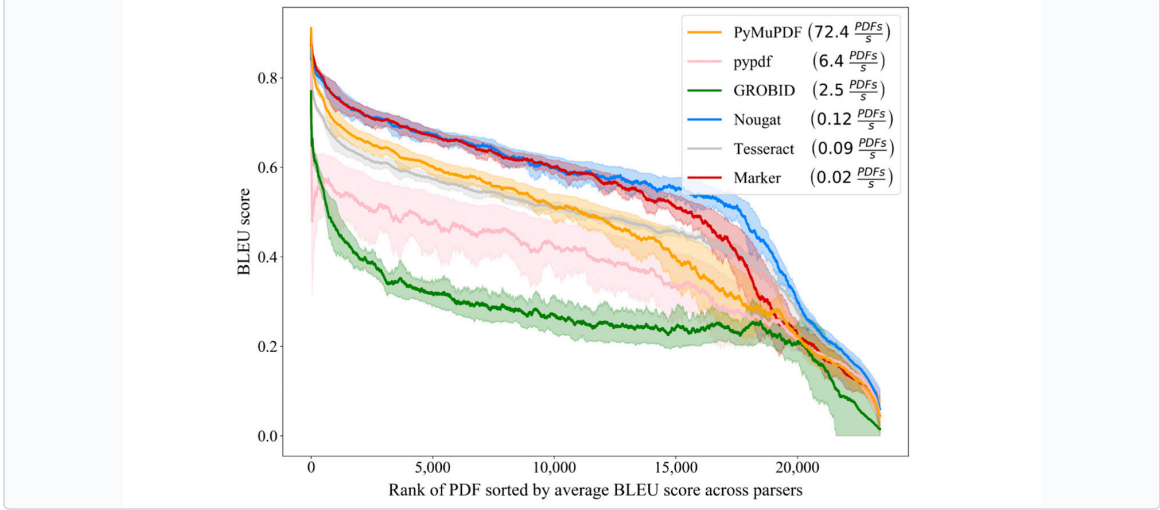

**MIST**, foundation models for chemistry and materials. *Polaris*




**Digital twins**, graph neural networks for advanced reactors. *Sophia*




**Automated inspection**, AI checking aerostructures during manufacture. *Polaris, Sophia*




**AdaParse**, millions of scientific PDFs into AI training data. *Polaris*

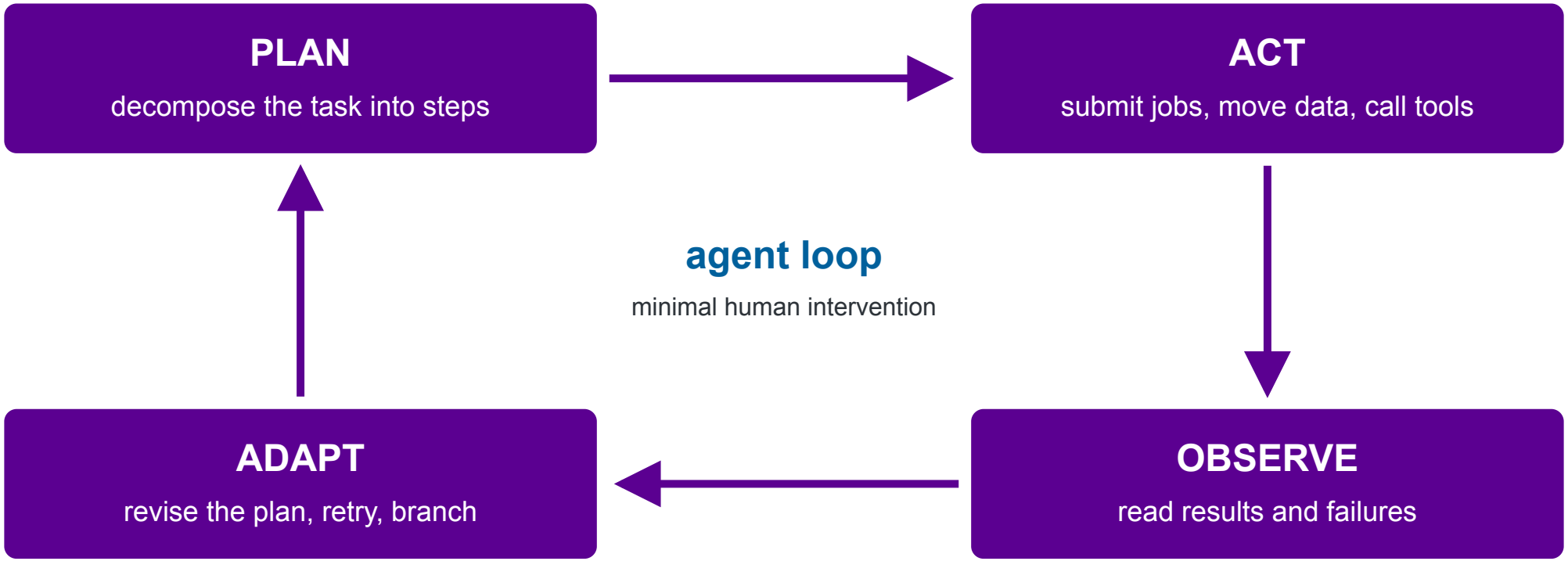
# AI Inference Service




served from dedicated inference systems, Sophia (NVIDIA DGX A100), Metis (SambaNova), Tara (Upcoming NVIDIA GH200)

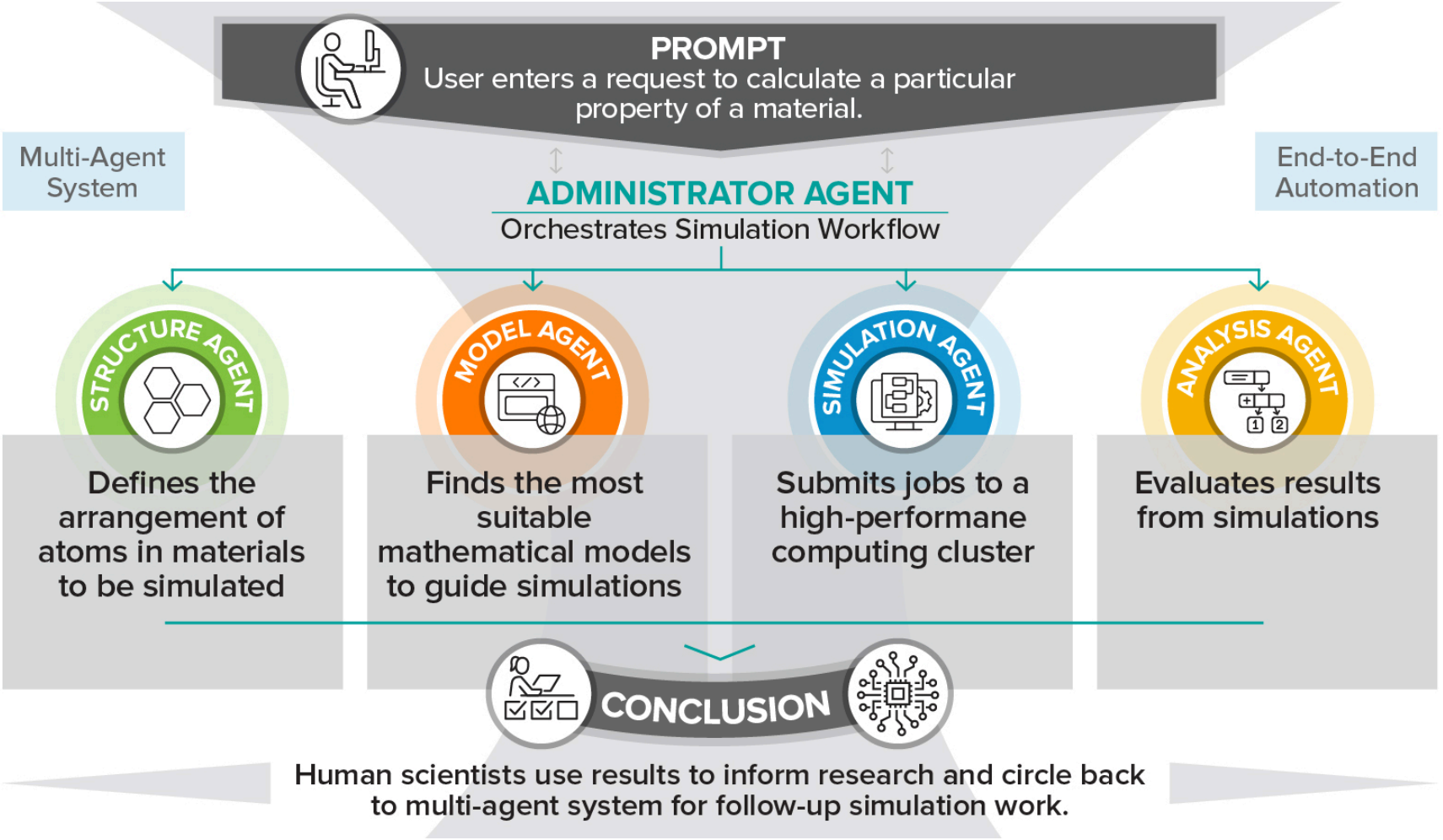
The **ALCF Inference Service** makes AI a shared facility resource, like storage or the network.

# The workflow layer is becoming agentic




ALCF: *"These workflows can reason, plan, and execute tasks based on new results with minimal human intervention."*

# Argonne is building agents that run experiments



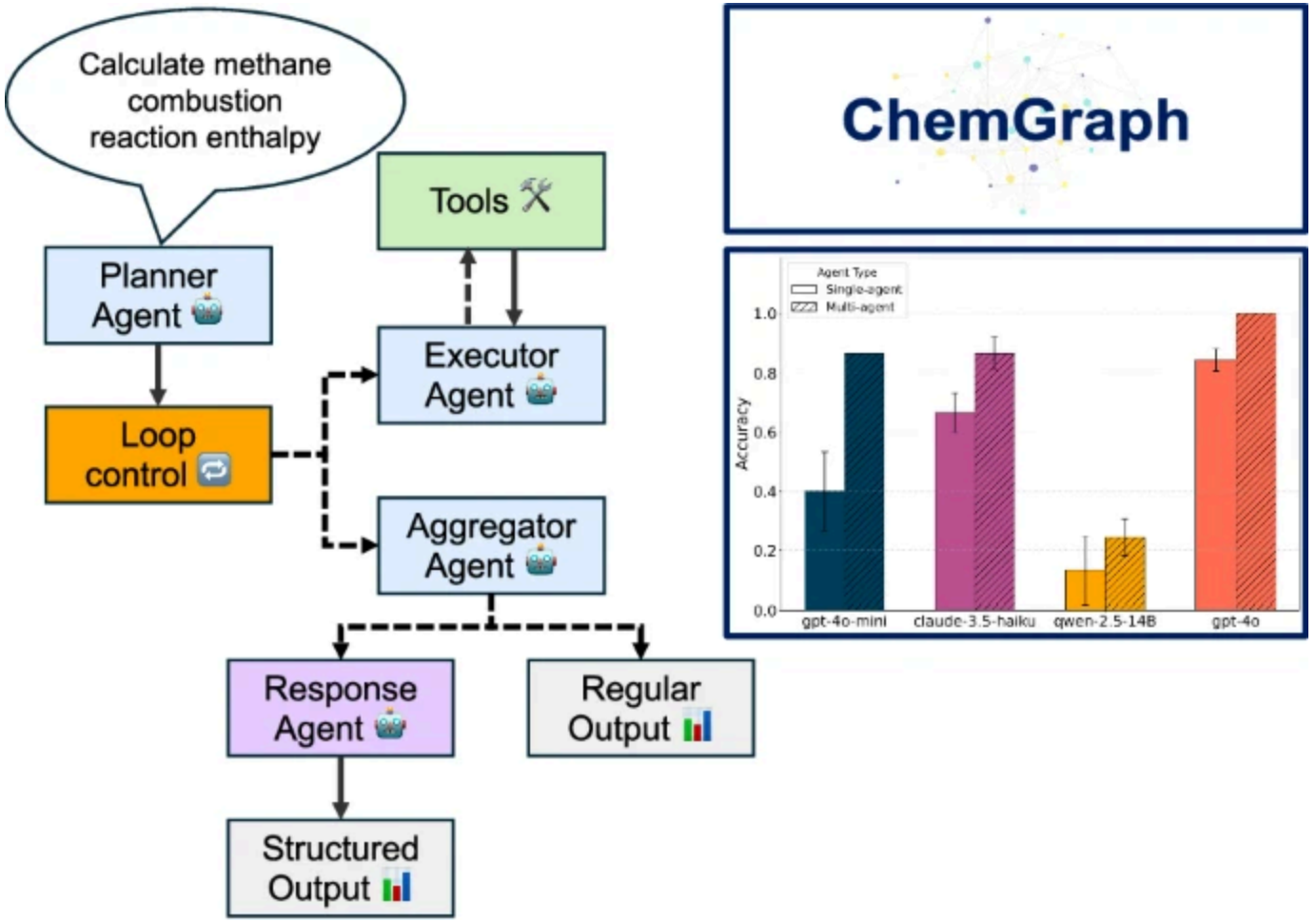

A multi-agent system for materials simulation: an administrator agent orchestrates structure, model, simulation and analysis agents

The simulation agent submits the jobs to a high-performance computing cluster

Alongside it, Argonne's autonomous discovery programme puts fixed and mobile robots in the laboratory itself

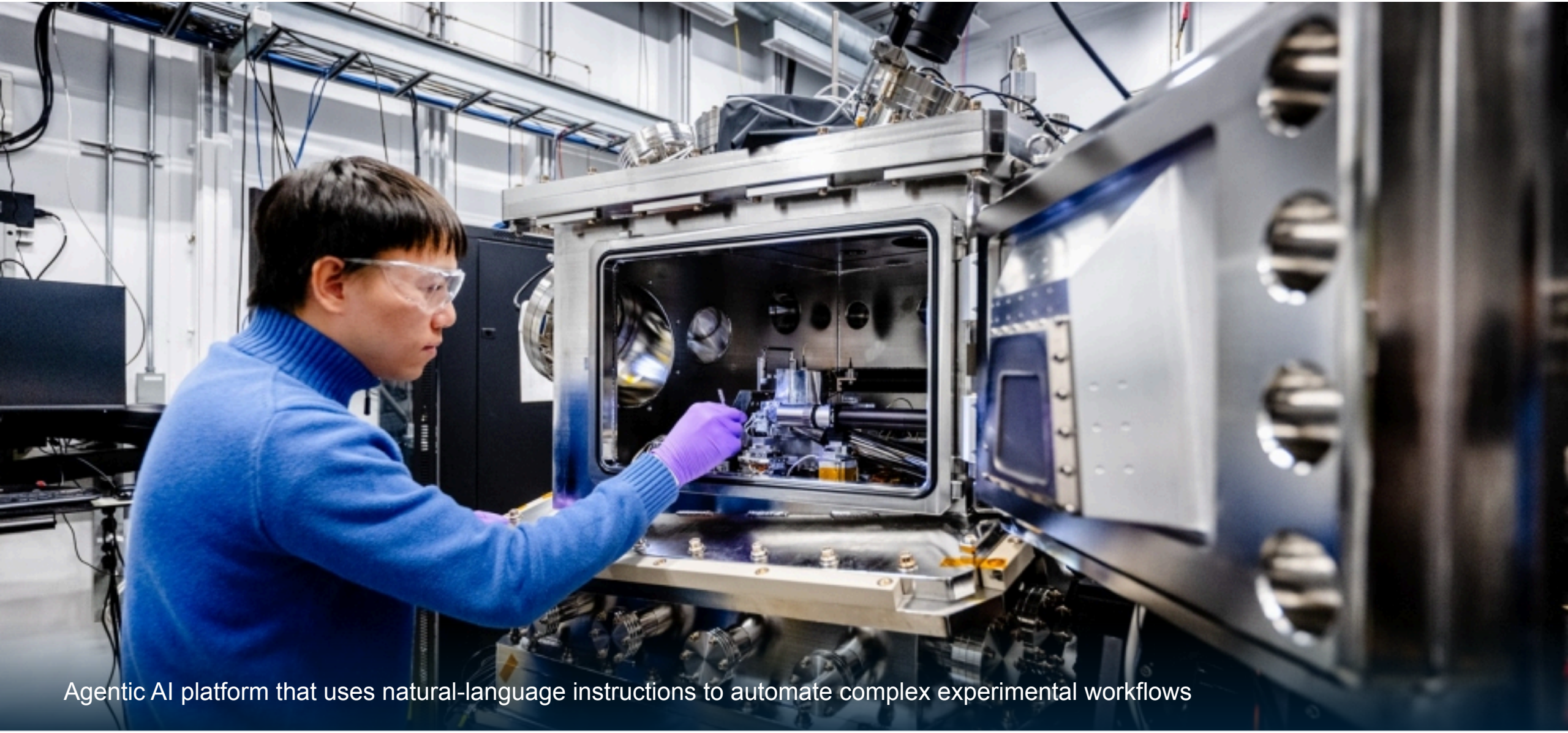
*Argonne news, Scientists deploy AI agents to accelerate discovery of new materials, 2026, and anl.gov/autonomous-discovery*

# ChemGraph agentic framework for computational chemistry




Opens quantum chemistry to non-experts, tracks AI model performance

# Agentic AI turns simple language into self-guided experimentation



Agentic AI platform that uses natural-language instructions to automate complex experimental workflows

*https://www.alcf.anl.gov/news/new-kind-microscope-agentic-ai-turns-simple-language-self-guided-experimentation*

05

# My Research

# Where I sit

## The science

X-ray imaging · graph analytics · streaming experiments

↓ ↑

## Declarative languages, system software and runtimes

my layer, let the science say *what*, and make the machine deliver it

↓ ↑

## HPC Systems

GPUs · nodes · interconnect · power

Four systems, one idea: raise the level the scientist writes at, without giving up the machine. **Monolog** (single GPU) › **MNMGDatalog** (many nodes) › **DRAVA** (streaming) › **Powerlog** (energy).

# Declarative analytics: say what, not how

Users expresses what to achieve with the data rather than how to accomplish it

**Employees**

| EmpID | Name | Email | Age | ManagerID |
|---|---|---|---|---|
| 0 | Alice | alice@example.com | 56 | Null |
| 1 | Bob | bob@example.com | 33 | 0 |
| 2 | Eve | eve@example.com | 26 | 0 |
| 3 | Carol | carol@example.com | 34 | 2 |
| 4 | Dave | dave@example.com | 22 | 3 |

Find all employees who are less than 30 years old



```
SELECT Name FROM Employees WHERE Age < 30;
```

Recursive queries, reachability, connected components, program analysis, in a few lines

The engine, not the scientist, decides how to run them on a GPU

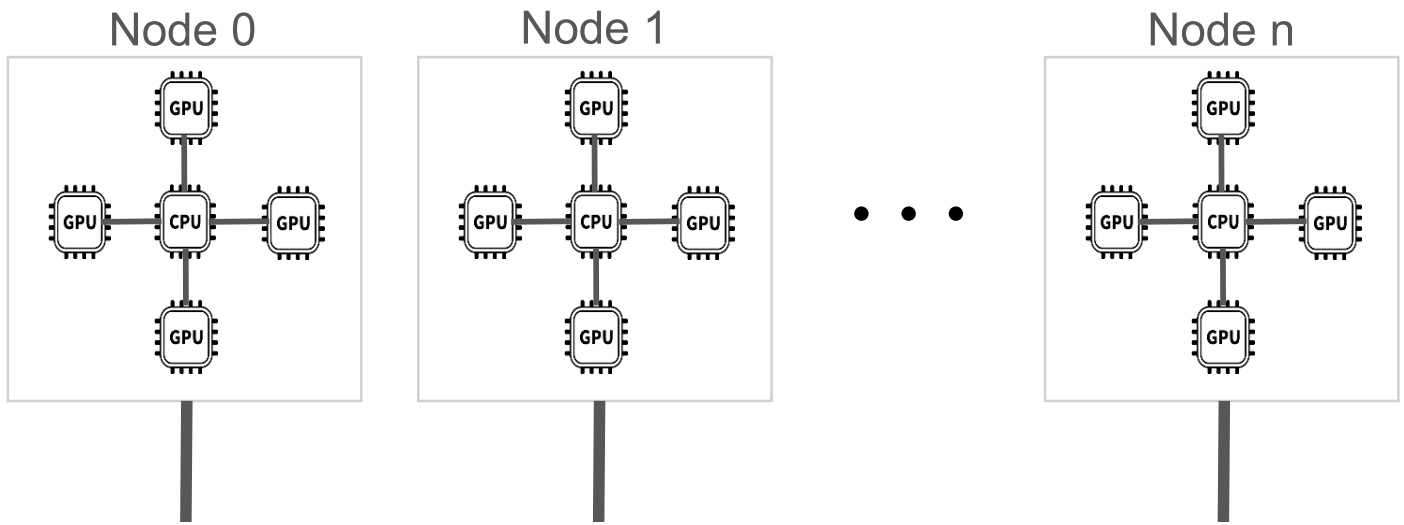
# High-Performance Deductive Database

Datalog

Relational algebra

Multi-node multi-GPU systems



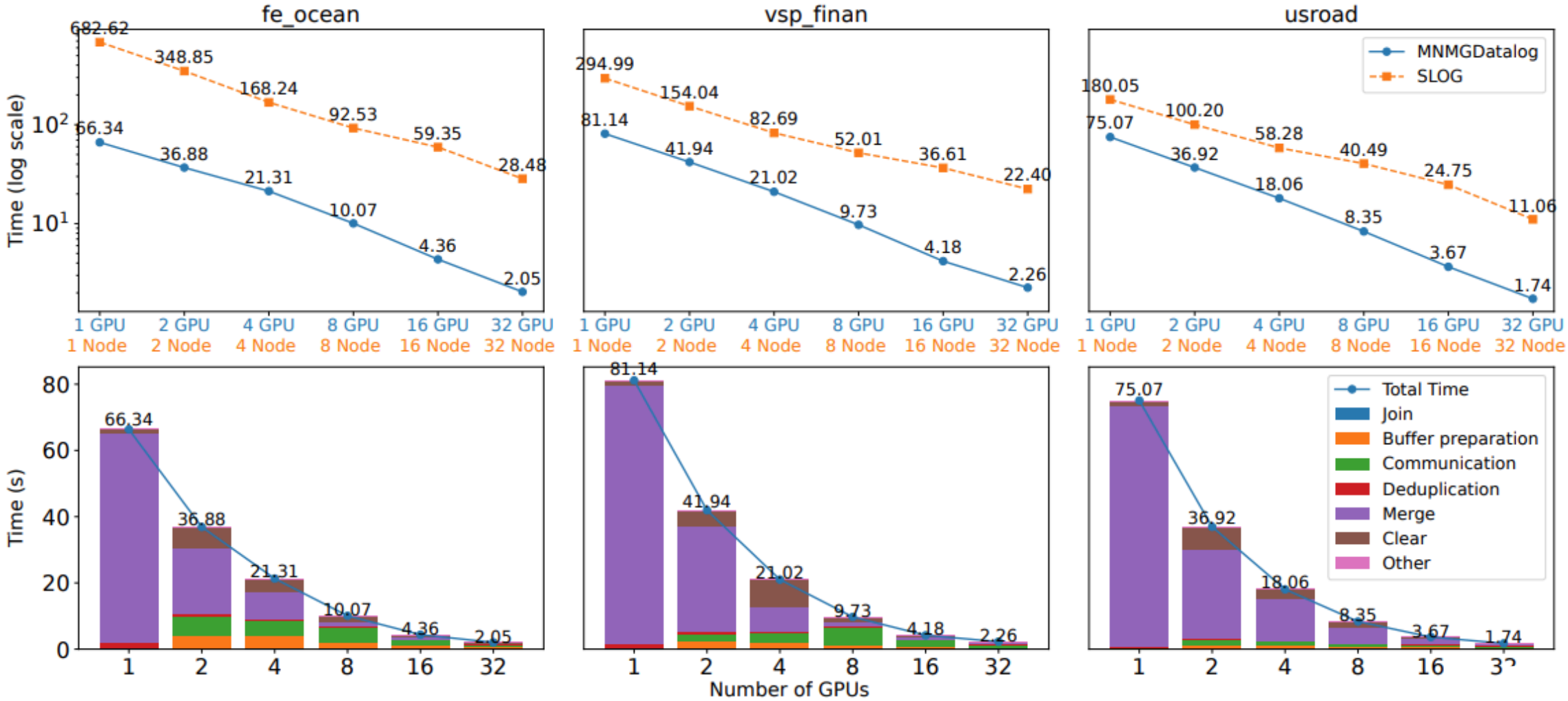




**High performance implementation for highly expressive language**

Expressive engines were CPU-only, fast GPU engines were single-device

Nothing scaled a recursive declarative engine across nodes

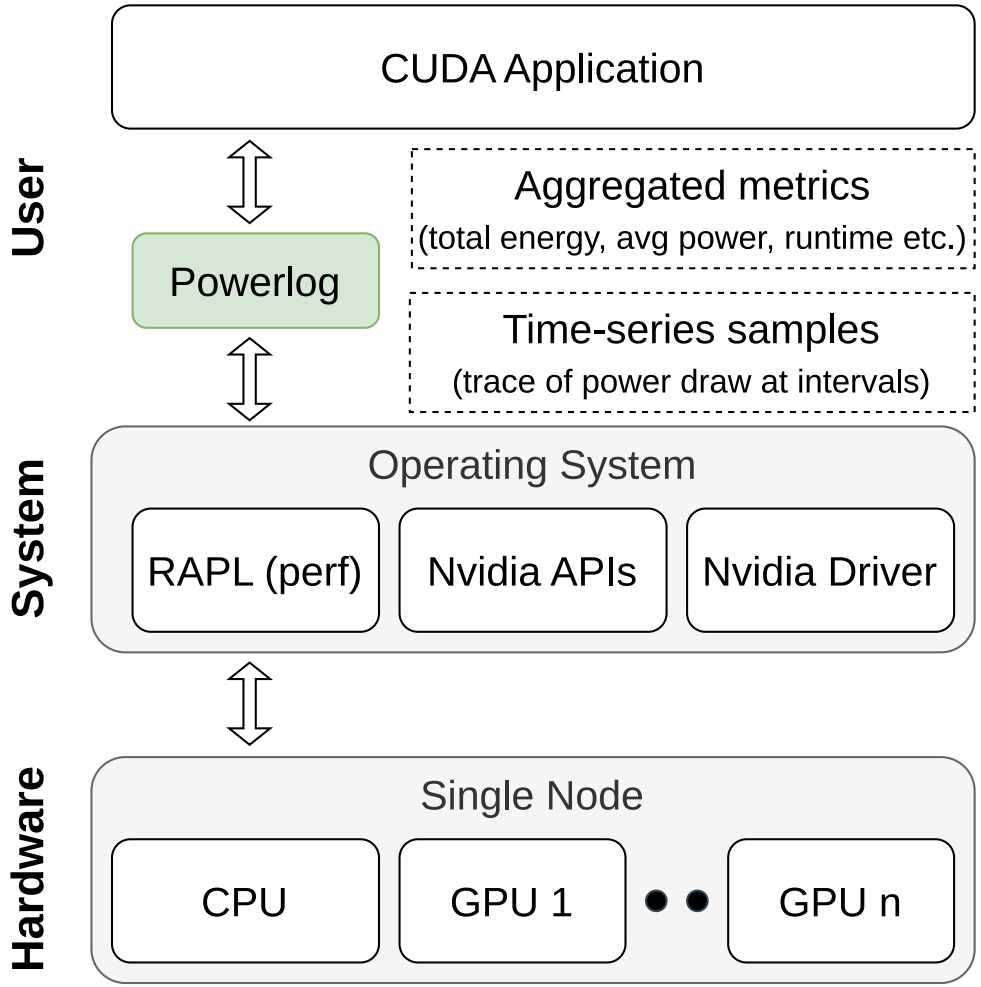
# The first multi-node multi-GPU Datalog engine




Transitive closure benchmark and breakdown up to 32 GPUs

Divide the graph analytics problem to multiple devices to analyze large scale graphs

# Heterogeneous power profiling (CPU<->GPU) using Powerlog

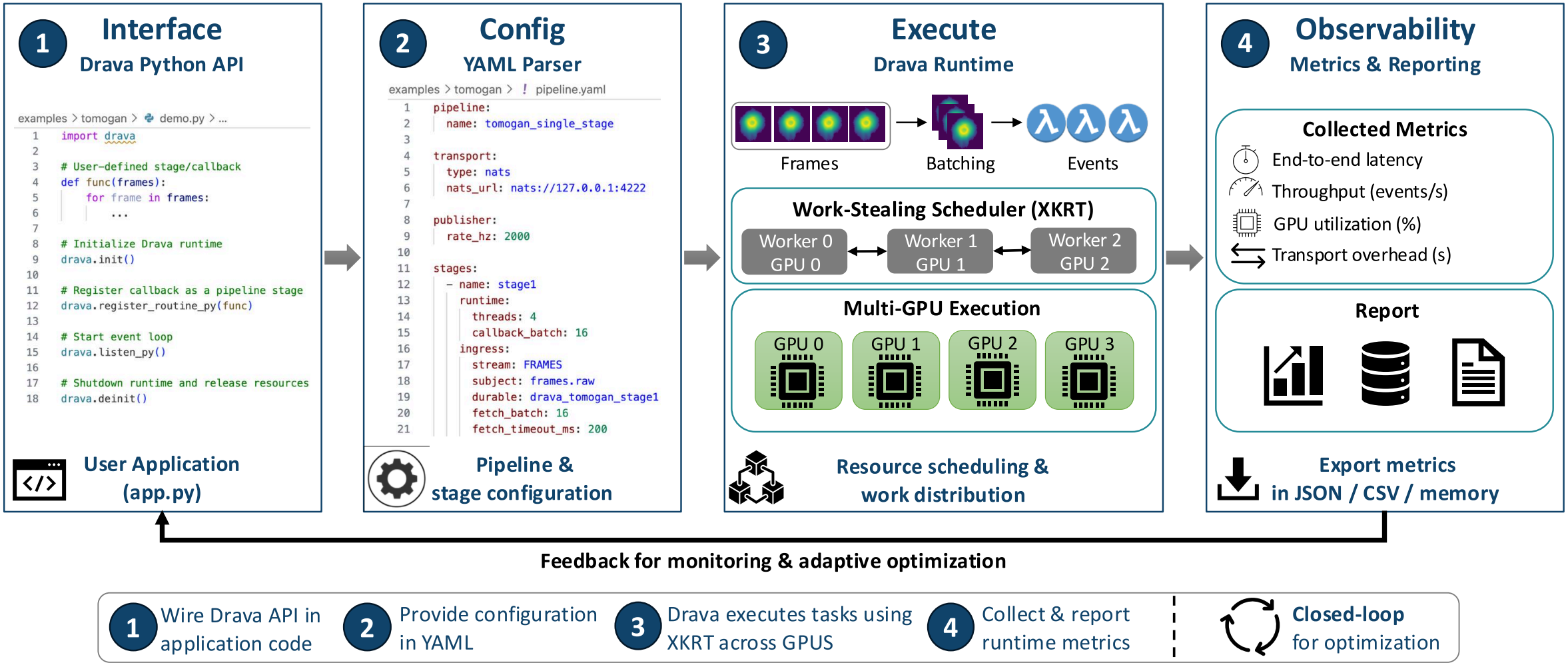



Samples CPU RAPL counters and the NVIDIA driver around a running kernel

Reports total energy, average power and runtime, plus a time series of power draw

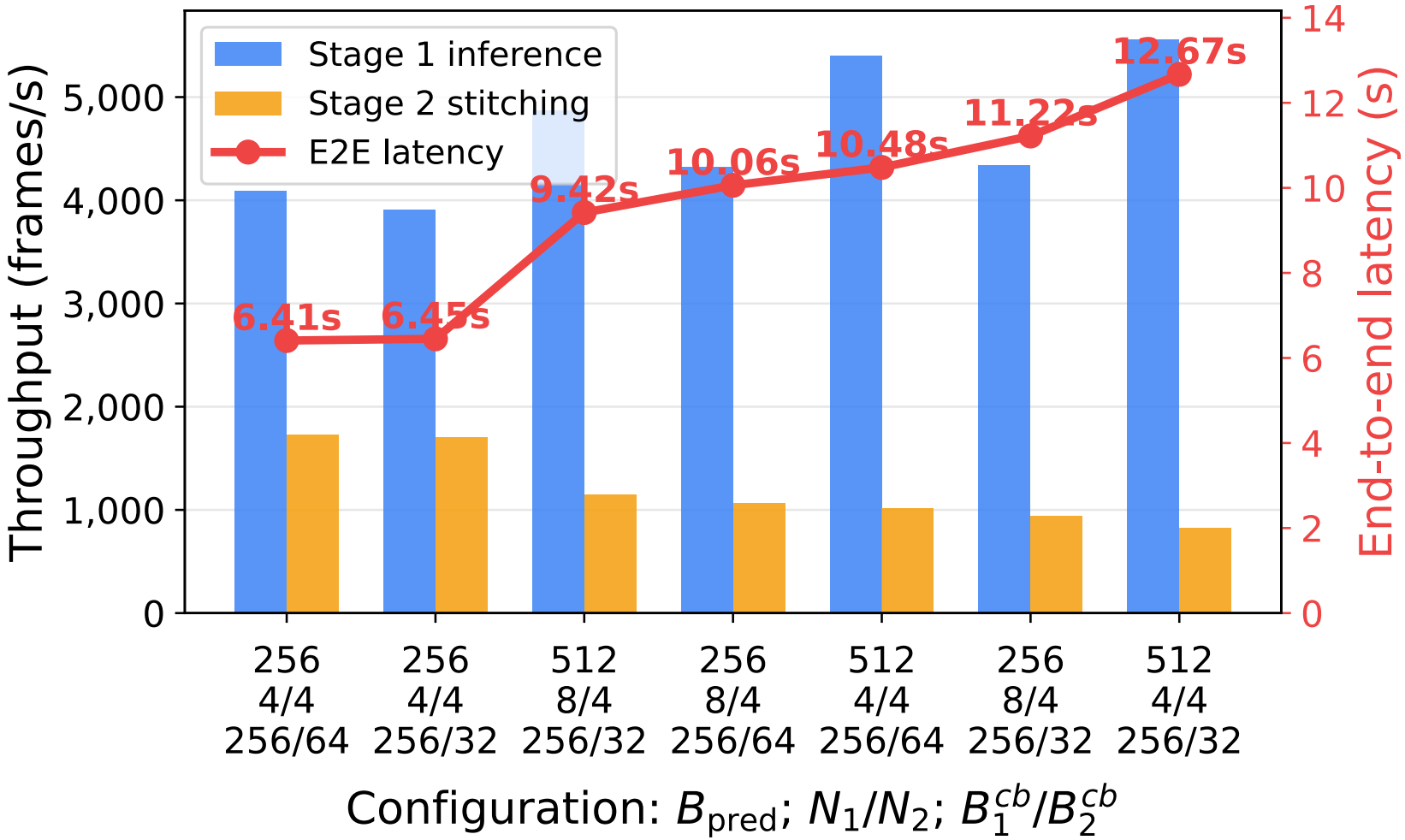
Open source, pip install powerlog

# DRAVA, a runtime for streaming experiments




Built so a beamline scientist configures a pipeline instead of writing one

# Manual tuning does not scale

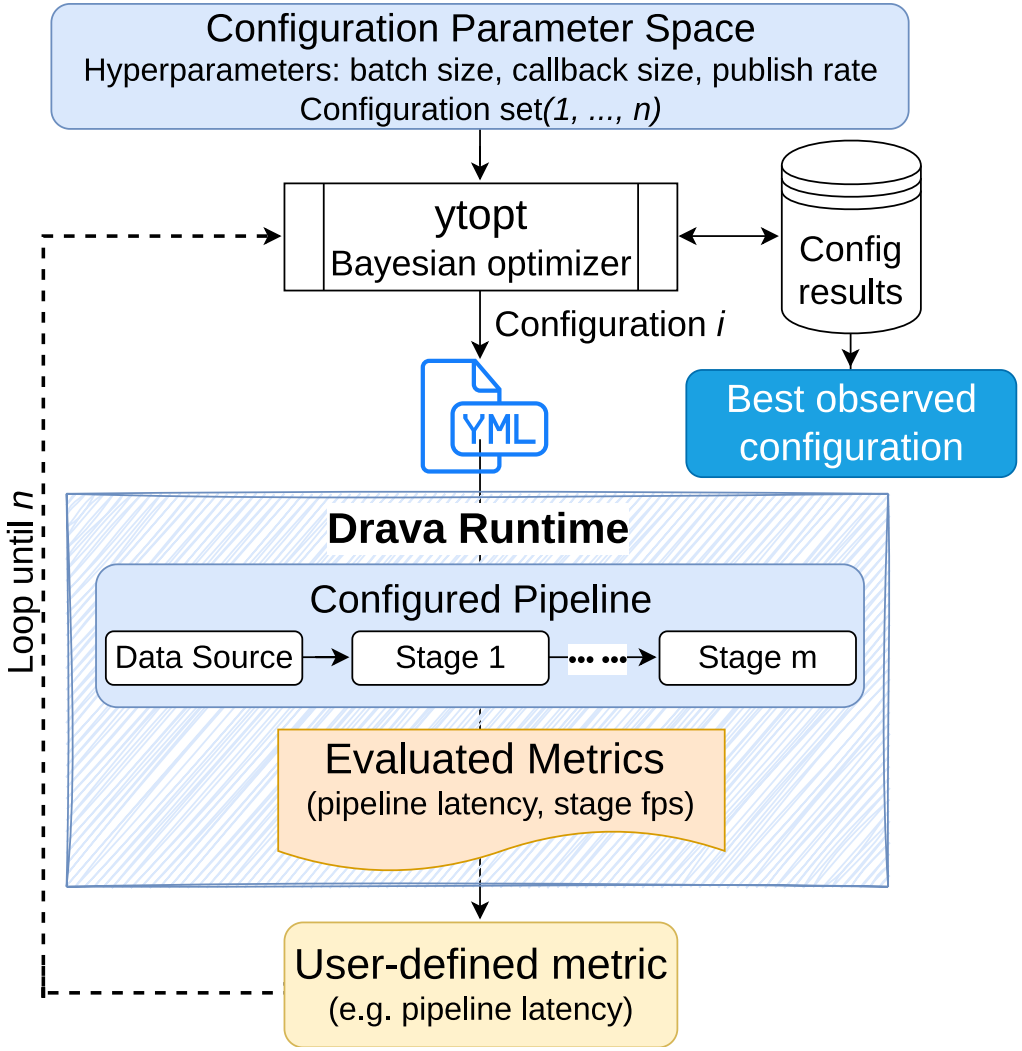



Seven hand-picked configurations of the same two-stage pipeline

The configuration with the fastest first stage is not the one with the best end-to-end latency

Batch size, transport, thread counts and placement all interact, so the space is combinatorial

# Agentic optimization to hypermater tuning




Batch size, transport, thread placement, callback grouping, a combinatorial configuration space

A Bayesian optimiser (ytopt) drives the runtime, measures end-to-end latency, and picks the next configuration

06

# Conclusion

# Take away

- HPC enables science at scales that are otherwise impossible
- AI is becoming a core part of modern HPC workloads and workflows
- Efficient runtimes and declarative languages make complex scientific computing easier
- The future is increasingly autonomous: simulation, AI, and agents working together

# Selected publications

## DECLARATIVE ANALYTICS ON GPUS

- **A. R. Shovon**, R. Pereira, Y. Sun, F. Saif, S. Kumar, S. Perarnau. *Accelerating GPU Datalog with Kernel Fusion.* IA³ Workshop, SC, **2026**.
- **A. R. Shovon**, Y. Sun, K. Micinski, T. Gilray, S. Kumar. *Multi-Node Multi-GPU Datalog.* ACM ICS, **2025**.
- Y. Sun, **A. R. Shovon**, T. Gilray, S. Kumar, K. Micinski. *Optimizing Datalog for the GPU.* ACM ASPLOS, **2025**.
- A. Sewell, K. Fan, **A. R. Shovon**, L. Dyken, S. Kumar, S. Petruzza. *Bruck Algorithm Performance Analysis for Multi-GPU All-to-All Communication.* HPC Asia, **2024**.
- **A. R. Shovon**, T. Gilray, K. Micinski, S. Kumar. *Towards Iterative Relational Algebra on the GPU.* USENIX ATC, **2023**.
- **A. R. Shovon**, L. R. Dyken, O. Green, T. Gilray, S. Kumar. *Accelerating Datalog Applications with cuDF.* IA³ Workshop, SC, **2022**.

## RUNTIMES AND ENERGY

- **A. R. Shovon**, R. Pereira, X. Wu, B. Videau, S. Perarnau. *DRAVA: An Event-Driven Runtime for Scientific Pipelines with Agentic Configuration Optimization.* XLOOP Workshop, SC, **2026**.
- **A. R. Shovon**, Y. Sun, Z. Lan, S. Perarnau, T. Gilray, K. Micinski, M. E. Papka, S. Kumar. *Heterogeneous Energy Characterization of GPU-Powered Datalog Engines.* EESP Workshop, SC, **2026**.

## CLOUD AND VISUALIZATION

- K. Balakavi, R. Janga, **A. R. Shovon**, D. Dempsey, E. Lefkowitz, S. Kumar. *Scalable, Interactive and Hierarchical Visualization of Virus Taxonomic Data.* IEEE VAHC at IEEE VIS, **2023**.
- S. Kumar, **A. R. Shovon**, G. Deshpande. *The Robustness of Persistent Homology of Brain Networks to Data Acquisition-Related Non-Neural Variability in Resting State fMRI.* Human Brain Mapping, **2023**.
- D. S. Chandrashekar, S. K. Karthikeyan, P. K. Korla, H. Patel, **A. R. Shovon**, et al. *UALCAN: An Update to the Integrated Cancer Data Analysis Platform.* Neoplasia, **2022**.
- M. Whaiduzzaman, A. Barros, **A. R. Shovon**, M. R. Hossain, C. Fidge. *A Resilient Fog-IoT Framework for Seamless Microservice Execution.* IEEE SCC, **2021**.
- M. Whaiduzzaman, M. R. Hossain, **A. R. Shovon**, S. Roy, A. Laszka, R. Buyya, A. Barros. *A Privacy-Preserving Mobile and Fog Computing Framework to Trace and Prevent COVID-19 Community Transmission.* IEEE Journal of Biomedical and Health Informatics, **2020**.
- **A. R. Shovon**, S. Roy, A. K. Shil, T. Atik. *GDPR Compliance: Implementation Use Cases for User Data Privacy in News Media Industry.* ICASERT, **2019**.
- **A. R. Shovon**, S. Roy, T. Sharma, M. Whaiduzzaman. *A RESTful E-Governance Application Framework for People Identity Verification in Cloud.* Springer LNCS, IEEE CLOUD, **2018**.
- S. Roy, **A. R. Shovon**, M. Whaiduzzaman. *Combined Approach of Tokenization and Mining to Secure and Optimize Big Data in Cloud Storage.* IEEE R10-HTC, **2017**.

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# Thank you

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